

BancaFiel Digital Transformation

AWS-Powered Loan & Credit Card Processing Solution

Detailed Technical & Financial Analysis

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Executive Summary

BancaFiel is losing \$1-2 million per month due to a manual loan and credit card application process that takes one week to complete. This delay causes 10% application loss, rising fraud from manual errors, and customer attrition to competitors who approve applications in hours.

Our proposed AWS solution automates 90% of the processing workflow—including document extraction, data validation, fraud detection, and workflow routing—while maintaining human oversight on all final credit decisions. This hybrid approach reduces processing time from 1 week to under 2 hours (97% reduction), cuts fraud by 70%, and costs only \$250-350 per month to operate.

The financial impact is substantial: \$741,000 to \$1.59 million in monthly savings through reduced fraud losses, recovered lost applications, and improved staff efficiency. With an annual ROI of 17,800% to 38,000%, the system pays for itself in less than one month.

Key Metrics at a Glance:

- Processing Time: 1 week → 2 hours (97% reduction)
- Application Loss: 10% → 3% (7% improvement = \$210K-525K/month recovered)
- Fraud Reduction: 70% decrease = \$525K-1M/month saved
- AWS Operating Cost: \$250-350/month
- Net Monthly Savings: \$741,000 - \$1,590,000
- Annual ROI: 17,800% - 38,000%

1. Problem Statement

1.1 Current Manual Process

BancaFiel currently processes loan and credit card applications through an eight-step manual workflow that creates significant operational bottlenecks:

Process Overview: Applications arrive via email with attached documents (INE/IFE, proof of address, bank statements). Staff manually download each attachment, open PDFs individually, extract customer data by hand, type information into Excel spreadsheets, search the bank database for validation, compare data across documents manually, route to approvers via email, wait for approval responses, and finally update the ERP system.

Current Performance Metrics:

- Volume: 500 applications per day (~15,000/month)
- Processing Time: Approximately 1 week per application
- Staff Required: 5-8 full-time employees for data entry and processing
- Application Loss Rate: 10% (1,500 applications/month)
- Manual Errors: Rising fraud incidents from data entry mistakes

1.2 Business Impact

The manual process creates four critical business problems:

Lost Revenue: The 10% application loss rate translates to 1,500 lost applications monthly. With an average loan value of \$10,000 and 2-5% profit margin, BancaFiel loses \$300,000 to \$750,000 per month in revenue opportunity. Customers who wait too long simply switch to competitors who approve applications in hours.

Fraud Losses: Manual data entry errors create vulnerabilities that fraudsters exploit. Estimated fraud rate of 1-2% results in 150-300 fraudulent approvals monthly, with average losses of \$5,000 per case. Total fraud losses: \$750,000 to \$1.5 million per month.

Competitive Disadvantage: Modern banks process applications in 2-4 hours using automated systems. BancaFiel's one-week timeline drives customers to competitors, particularly younger, tech-savvy customers who expect instant digital experiences.

Staff Burnout: Employees spend 30-60 minutes per application on repetitive data entry tasks. This tedious work causes high turnover, low morale, and prevents staff from focusing on high-value activities like customer relationship building and complex case resolution.

2. Proposed Solution

2.1 Solution Approach: Human + AI

Our solution follows a conservative, banking-appropriate design: automate 90% of the work, but humans make all final credit decisions.

Rationale: We don't yet know BancaFiel's specific approval criteria (income thresholds, credit score requirements, debt-to-income ratios). These rules will be gathered in future sessions with KPMG. Additionally, banking regulations may require human oversight for regulatory compliance. This conservative approach builds trust with executives while still delivering massive time savings.

What Gets Automated:

- Document Receipt & Storage (Amazon S3, SES)
- AI-Powered Data Extraction (Amazon Textract)
- Data Validation Against Customer Database (AWS Lambda + RDS)
- Fraud Risk Scoring (Amazon Fraud Detector)
- Workflow Routing to Approvers (Step Functions)
- ERP System Updates (AWS Lambda)
- Customer Notifications (Amazon SES)

What Stays Manual:

- Final Approve/Reject Decision (Bank Staff)
- Review of AI Recommendations and Risk Scores
- Edge Case Handling and Complex Scenario Resolution
- Regulatory Compliance Oversight

2.2 Approver Experience Transformation

The solution transforms the approver's role from data entry clerk to decision maker.

Before (Manual): Download 5 PDFs → Read each document → Extract data manually → Type into Excel → Search bank database → Compare data across documents → Check for inconsistencies → Make decision → Reply to email manually. Time: 30-60 minutes per application.

After (AWS): Open web dashboard → Review pre-extracted data → See fraud risk score (Green/Yellow/Red) → View AI recommendation → Check linked documents if needed → Click Approve or Reject → System automatically updates ERP and notifies customer. Time: 30 seconds to 5 minutes per application.

2.3 Core AWS Services

The solution leverages six core AWS services:

Amazon Textract: AI-powered document extraction. Automatically reads PDFs and extracts structured data (names, addresses, ID numbers) from INE/IFE, proof of address, and bank statements. No manual data entry required.

AWS Lambda: Serverless compute functions that process documents, validate data against the database, and integrate with the ERP system. Scales automatically with application volume.

Amazon RDS: Managed database (PostgreSQL/MySQL) that stores customer data, application history, and audit logs. Replaces Excel spreadsheets with a proper relational database.

Step Functions: Visual workflow orchestration that manages the entire process from document upload to final decision. Tracks application status, handles errors automatically, and provides audit trail.

Amazon Fraud Detector: Machine learning service that analyzes applications for fraud risk. Detects duplicate applications, inconsistent data, and suspicious patterns. Assigns risk scores (0-1000) to flag high-risk cases.

API Gateway + Amplify: Web application infrastructure for customer uploads and staff approval dashboard. Secure, scalable, and integrated with authentication (Amazon Cognito).

3. Cost Analysis

3.1 AWS Operating Costs

Production environment monthly costs for 15,000 applications per month:

AWS Service	Purpose	Monthly Cost
Amazon Textract	Document extraction	\$135
Amazon RDS	Database	\$36
Amazon Fraud Detector	Fraud prevention	\$23
AWS Lambda	Compute	\$2
Amazon S3	Storage	\$8
AWS Step Functions	Workflow	\$4
Amazon CloudWatch	Monitoring	\$15
Amazon SES	Email	\$1
API Gateway	API	\$1
Amazon Cognito	Authentication	\$0
TOTAL		\$250-350

Total AWS Operating Cost: \$250-350 per month

Note: Most services fall within AWS Free Tier for first 12 months. Student credits available through AWS Educate (\$100/year) cover demo development costs.

3.2 Current Business Costs

BancaFiel's current manual process generates the following monthly costs and losses:

Cost Category	Monthly Cost/Loss
Staff Labor (5-8 employees)	\$10,000 - \$24,000

Lost Applications (10% loss rate)	\$300,000 - \$750,000
Fraud Losses (1-2% fraud rate)	\$750,000 - \$1,500,000
Customer Churn	Significant (unquantified)
TOTAL	\$1,060,000 - \$2,274,000

Total Current Monthly Costs: \$1,060,000 - \$2,274,000

3.3 ROI Analysis

The AWS solution delivers substantial financial returns:

Category	Current	With AWS
IT Operations	\$0	\$300
Staff Labor	\$10K-24K	\$4K-6K
Lost Applications	\$300K-750K	\$90K-225K
Fraud Losses	\$750K-1.5M	\$225K-450K
Monthly Savings		\$741K-1.59M

Key Financial Metrics:

- Net Monthly Savings: \$741,000 - \$1,590,000
- Annual Savings: \$8.9M - \$19M
- Annual ROI: 17,800% - 38,000%
- Payback Period: Less than 1 month

4. Conclusion

BancaFiel's manual loan processing system is costing \$1-2 million per month in fraud losses, lost applications, and operational inefficiency. The proposed AWS solution automates 90% of the workflow while maintaining essential human oversight on credit decisions, delivering a 97% reduction in processing time at a negligible operating cost of \$250-350 per month.

With monthly savings of \$741,000 to \$1.59 million and an ROI exceeding 17,800% annually, this is not just a technology upgrade—it's a critical business imperative. The system pays for

itself in less than one month and positions BancaFiel to compete effectively with modern banks that have already automated their operations.

The real question is not whether BancaFiel can afford to implement this solution, but whether they can afford not to.

Recommended Next Steps:

1. Requirements Gathering Session: Confirm approval criteria and ERP integration details
2. Prototype Demonstration: Working demo of document extraction with Textract
3. Architecture Review: Final approval of AWS services and cost estimates
4. 4-Week Implementation: Build, test, train staff, and deploy to production